Haotian Liu

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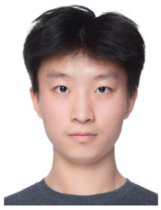

EDUCATION

Xiamen University (XMU), Xiamen, China Sep. 2023 – Jun. 2027 (expected)

B.Eng. in Software Engineering | Overall Average: 82.50/100

Relevant Coursework: Machine Learning for Decision-Making and Optimization (96); Algorithms for Modern AI (94); Introduction to Artificial Intelligence (93); Interactive Design (90); Data Warehouse (90).

PUBLICATIONS

PilotBench: A Benchmark for General Aviation Agents with Safety Constraints.
IJCNN 2026, **accepted**. Co-first author. arXiv:2604.08987

Long-Horizon Memory Governance for LLM Agents.
Under review at EMNLP 2026. First author.

ProcuraClaw: Context-Aware Procurement Search via Adaptive Hotel Profiling and Centralization Rate Prediction.
Under review at CIKM 2026. Co-first author.

RESEARCH PROJECTS

DDRG: Reasoning-Graph Verification and Repair Apr. 2026 – Present

Research Collaborator | Li Lab

- Prototyped an inference-time reasoning-graph pipeline that samples candidate graphs, probes answer-sensitive claims, and gates local repairs conservatively.
- Evaluated Llama 3.1 70B across multiple reasoning benchmarks; case-level audits characterized parser, verifier, and repair-selection failures and informed conservative answer-selection gates.

ConcordCoder: Pre-generation Alignment for AI Coding Mar. 2026 – Present

Online Research Intern | Waseda Institute for Advanced Study

- To prevent intent and compatibility errors before AI-generated patches, proposed **ConcordCoder**, which reconstructs code-grounded task context and asks developers to confirm constraints, risks, and non-goals before generation.
- Designed an inspectable **AlignmentRecord** and a controlled evaluation of comprehension, review accuracy, evidence grounding, guard violations, and multi-round constraint retention.

NeuroMark: Multimodal Clinical Reasoning Sep. 2025 – Present

Research Contributor | Remote Research Collaboration | School of Data Science, Fudan University

- Led the initial project phase and evaluated 20 LLMs on 2,379 clinical questions and 16 VLMs across six brain-imaging tasks.
- Contributed to reliability analyses using safety compliance, effective accuracy, and difficulty discrimination; paired-model results showed no significant association between text and imaging performance ($\rho = 0.14$, $p = 0.76$).

Long-Horizon Memory Governance for LLM Agents (LHMG) Dec. 2025 – May 2026

Project Lead | Remote Research Collaboration | School of Data Science, Fudan University

- Developed a prototype memory-governance layer combining append-only revisions, active-state semantics, risk-gated conditioning, and governed forgetting for auditable long-horizon agent memory.
- In controlled diagnostics, observed a 13.3-point GPT-4o gain in cross-turn decision consistency over flat memory.

PilotBench: Safety-Critical Aviation Agents Dec. 2025 – Mar. 2026

Core Contributor | Remote Research Collaboration | School of Data Science, Fudan University

- Co-developed PilotBench from 708 real trajectories, 34-channel telemetry, and nine flight phases; contributed to evaluations of 41 models.
- Contributed to Pilot-Score design and analyses identifying a precision–controllability dichotomy and a dynamic-complexity gap in high-workload phases.

INDUSTRY EXPERIENCE

ProcuraClaw: Context-Aware Procurement Search Nov. 2025 – Apr. 2026

AI Research Intern | H World Group

- Contributed to a deployed procurement-search system using LLM entity extraction, hybrid retrieval, profile-aware ranking, and Thompson Sampling.
- The system achieved MRR 0.95 and NDCG@5 0.91 on 175 evaluation queries; reported live A/B tests increased first-hit CTR by 6.2% and 9.8% across two deployment slices.

TECHNICAL SKILLS AND LANGUAGES

Programming and ML: Python, PyTorch, Transformers, SQL, Java; LLM agents, RAG, multimodal learning, evaluation, and information retrieval.
Research Tools: Git, Linux, Docker, LaTeX; experiment design, ablation studies, reproducible evaluation, and academic writing.
Languages and Tests: Chinese (native); English (TOEFL iBT 90); Japanese (working reading proficiency).